

TOPIC NAME: Fragmentation

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Summer 2025 (A)

② i)

Header size = 38 B

Packet size = Data + Header

At 10th fragment, offset = 1854

MF = 1

So, before it 9 fragments have done

$$\text{Offset} = \frac{\text{total data proceed}}{8} \times 9$$

$$\Rightarrow \text{Data proceed} = \frac{1854 \times 8}{9} = 1648 \text{ B}$$

$$\text{Packet size} = 1648 + 38 = 1686 \text{ B}$$

Alternative:

Note: offset represents the starting position of the data in 8 byte units and the MF flag indicates if more fragments are following.

$$\text{Fragment 10 start} = 1854 \times 8 = 14832 \text{ B}$$

$$\text{Fragment 15 start} = 2884 \times 8 = 23072 \text{ B}$$

$$\text{Total distance between 10 and 15} = 23072 - 14832 = 8240 \text{ B}$$

$$\text{Number of fragments} = 15 - 10 = 5$$

$$\text{Data length per fragment} = 8240 / 5 = 1648 \text{ B}$$

$$\text{Packet size} = \text{data length} + \text{header size} = 1648 + 38 = 1686 \text{ B}$$

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ii) Last Fragment starting position = 2884×8
 $= 23072 \text{ B}$

Last fragment data length = 69 B

Total data size = $23072 + 69 = 23141$ bytes

iii) Because last 3 bit position is considered as 000. So dividing by 8 on right shift 3 bit prevent any data loss.

Spring 25 (A)

③ i) As the 9th packet size = 272 bytes and MF=1. That means the packet must be in full size and more fragment to come.

MTU = 272 bytes.

ii) Total size = 272

Header = 32

Data = $272 - 32 = 240$

Before 7th packet, there are 6 packets.

So, total preceding byte = $240 \times 6 = 1440$ byte

Offset = $1440 / 8 = 180$

The fragment offset of the 7th packet = 180

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iii) Original Packet size = 2584

Header = 32

Total Data = $2584 - 32 = 2552$ bytes

240 byte data can fit per full segment.

$$\begin{array}{r} 240 \times 10 = 2400 \\ 2552 - 2400 = 152 \end{array}$$

So, number of full fragments = 10

Since there is leftover data (152 bytes), we need

one more fragment to carry it.

Total fragment = $10 + 1 = 11$

Fall 2024 (A)

③

Total length = 5086 bytes

Header length (IHL=5) : $5 \times 4 = 20$ bytes

Payload Data = $5086 - 20 = 5066$ bytes

MTU = 1244 bytes

Data per fragment = $1244 - 20 = 1224$ bytes

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i)

$$\text{Number of fragments} = \lceil 5066 / 1224 \rceil$$
$$= \lceil 4.138 \rceil = 5$$

ii) Data in first 4 fragments = 4×1224

$$= 4896 \text{ bytes}$$

$$\text{Data in last fragment} = 5066 - 4896 = 170 \text{ bytes}$$

$$\text{Total size of last packet} = 170 + 20 = 190 \text{ bytes}$$

iii) Data send before 5th fragment = $(4 \times 1224) \text{ B}$

$$= 4896 \text{ B}$$

$$\text{Fragment offset} = \frac{4896}{8} = 612$$

iv) Identification field (IF) allows the destination to recognize which fragments belong together to reassemble the original data correctly.

v) DF = 1 means do not fragment. So router can not fragment the oversize packet so it will discard the packet and send an error message back to sender.

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Summer 2024 (A)

③ i) Packet size = 7240 B

Header size = 40 B

MTU = 800 B

Data = 800 - 40 = 760 B

Total data size = 7240 - 40 = 7200 B

$$\text{Fragment} = \left\lceil \frac{7200}{760} \right\rceil = 9.47 \approx 10$$

ii) Data after 9th fragment = $\frac{7200 - (760 \times 9)}{1}$ (ii)
= 360 B

$$\text{Fragment size} = \frac{360}{1} + 40 = 400 \text{ B}$$

(iii) The fragment offset of the 8th fragment (iii)

$$= \frac{\text{total data up to 7th fragment}}{8}$$

$$= \frac{760 \times 7}{8} = 665$$

(iv) MF = 0 means no more segment ~~been done~~ left, as this is the very last piece of data.

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Spring 2024 (A)

③ 1) Data size (Payload) = 4080 bytes

MTU = 540 bytes

header size = 20 bytes

Max payload per fragment = $540 - 20 = 520$ B

Number of fragments = $\left\lceil \frac{4080}{520} \right\rceil = \lceil 7.85 \rceil = 8$

(ii) Fragment offset of the 4th fragment

$$= \frac{3 \times 520}{8} = 195$$

(iii) The 4th fragment is not the last fragment

Therefore, MF = 1